

Stack Overflow Developer Insights Survey

SQL Analysis Project Documentation

Technologies Used: PostgreSQL, pgAdmin 4

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1. Project Overview

This project performs a structured SQL-based analysis of the Stack Overflow Developer Survey dataset. It explores compensation trends across programming languages, databases, cloud platforms, countries, and work experience levels, providing data-driven insights into the global developer job market.

The analysis is implemented entirely in PostgreSQL and demonstrates core SQL techniques including window functions, array operations, CTEs, and conditional aggregation.

1.1 Objectives

- Calculate global and segmented median developer salaries
- Identify highest-paying programming languages, databases, and platforms
- Understand how work experience correlates with compensation
- Compare salaries across remote, hybrid, and in-person work arrangements
- Perform geographic salary analysis by country

1.2 Tools & Technologies

Component	Technology
Database Engine	PostgreSQL 12+
Query Language	SQL (ANSI + PostgreSQL extensions)
Dataset Format	CSV (~49,200 rows)
Key Extension	PERCENTILE_CONT, UNNEST, STRING_TO_ARRAY

2. Dataset Description

The dataset is sourced from Stack Overflow's Annual Developer Survey. Each row represents a single developer respondent. The table below describes each field:

Column Name	Data Type	Description
ResponseId	TEXT	Unique identifier per survey respondent
Country	TEXT	Country of residence of the developer
WorkExp	TEXT (numeric)	Years of professional work experience
RemoteWork	TEXT	Work arrangement: Remote, Hybrid, or In-person
ConvertedCompYearly	TEXT (numeric)	Annual compensation converted to USD
LanguageHaveWorkedWith	TEXT	Semicolon-separated list of programming languages used
DatabaseHaveWorkedWith	TEXT	Semicolon-separated list of databases used
PlatformHaveWorkedWith	TEXT	Semicolon-separated list of platforms/tools used

Note: Null/missing numeric values are stored as the string 'nan' and are explicitly excluded from all salary calculations using `WHERE ConvertedCompYearly <> 'nan'`.

3. Database Schema

The data is loaded into a single flat table:

```
DROP TABLE IF EXISTS developer_survey;

CREATE TABLE developer_survey (
    ResponseId          TEXT,
    Country              TEXT,
    WorkExp              TEXT,
    RemoteWork           TEXT,
    ConvertedCompYearly  TEXT,
    LanguageHaveWorkedWith TEXT,
    DatabaseHaveWorkedWith TEXT,
    PlatformHaveWorkedWith TEXT
);
```

All columns are stored as TEXT to accommodate missing values ('nan') and semicolon-delimited multi-value fields, with type casting applied at query time.

4. SQL Query Analysis

4.1 Global Median Salary

This query computes the worldwide median annual compensation across all developers who reported a valid salary.

```
SELECT
  ROUND(
    PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY ConvertedCompYearly::NUMERIC)
  ) AS Global_Median_Salary
FROM developer_survey
WHERE ConvertedCompYearly <> 'nan';
```

Key technique: `PERCENTILE_CONT(0.5)` is PostgreSQL's continuous percentile function that computes the exact median using linear interpolation.

4.2 Salary by Work Experience

Developers are grouped into five experience bands using a `CASE WHEN` expression, and median salary is calculated for each band.

```
SELECT
  CASE
    WHEN WorkExp::INT <= 2 THEN '0-2 Years'
    WHEN WorkExp::INT <= 5 THEN '3-5 Years'
    WHEN WorkExp::INT <= 10 THEN '6-10 Years'
    WHEN WorkExp::INT <= 15 THEN '11-15 Years'
    ELSE '15+ Years'
  END AS Experience_Level,
  ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
    ConvertedCompYearly::NUMERIC)) AS Median_Salary
FROM developer_survey
WHERE WorkExp <> 'nan' AND ConvertedCompYearly <> 'nan'
GROUP BY Experience_Level
ORDER BY Median_Salary DESC;
```

4.3 Salary by Country

Cross-tabulates median salary by country, filtered to countries with more than 50 respondents to ensure statistical reliability.

```
SELECT Country,
       ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
ConvertedCompYearly::NUMERIC)) AS Median_Salary
FROM developer_survey
WHERE ConvertedCompYearly <> 'nan'
GROUP BY Country
HAVING COUNT(*) > 50
ORDER BY Median_Salary DESC;
```

4.4 Salary by Remote Work Status

Compares median compensation across the three work arrangements: Remote, Hybrid, and In-person.

```
SELECT RemoteWork,
       ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
ConvertedCompYearly::NUMERIC)) AS Median_Salary
FROM developer_survey
WHERE ConvertedCompYearly <> 'nan'
GROUP BY RemoteWork
ORDER BY Median_Salary DESC;
```

4.5 Multi-Value Field Parsing: UNNEST & STRING_TO_ARRAY

Several survey fields (languages, databases, platforms) store multiple values in a single semicolon-delimited string. These are parsed using `STRING_TO_ARRAY` and exploded into individual rows using `UNNEST`.

Array conversion example:

```
SELECT ResponseId,
       STRING_TO_ARRAY(LanguageHaveWorkedWith, ';') AS Language_Array
FROM developer_survey LIMIT 10;
```

Unnested (one row per language):

```
SELECT ResponseId,
       UNNEST(STRING_TO_ARRAY(LanguageHaveWorkedWith, ';')) AS Language
FROM developer_survey LIMIT 20;
```

4.6 Highest Paying Programming Languages

Uses a CTE to unnest languages, then aggregates median salary per language (minimum 100 developers required).

```
WITH Languages AS (  
    SELECT ConvertedCompYearly::NUMERIC AS Salary,  
           UNNEST(STRING_TO_ARRAY(LanguageHaveWorkedWith, ';')) AS Language  
    FROM developer_survey  
    WHERE LanguageHaveWorkedWith IS NOT NULL AND ConvertedCompYearly <> 'nan'  
)  
SELECT Language, COUNT(*) AS Developers,  
       ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY Salary)) AS Median_Salary  
FROM Languages  
GROUP BY Language  
HAVING COUNT(*) > 100  
ORDER BY Median_Salary DESC;
```

4.7 Highest Paying Databases & Platforms

The same CTE pattern is applied to DatabaseHaveWorkedWith and PlatformHaveWorkedWith to identify which database technologies and deployment platforms are associated with higher compensation.

4.8 Final Project Query: Language x Experience Analysis

The most complex query combines language unnesting with experience-level bucketing to produce a cross-dimensional view of median salary by both language and experience.

```
WITH Languages AS (  
    SELECT WorkExp, ConvertedCompYearly::NUMERIC AS Salary,  
           UNNEST(STRING_TO_ARRAY(LanguageHaveWorkedWith, ';')) AS Language  
    FROM developer_survey  
    WHERE WorkExp <> 'nan' AND ConvertedCompYearly <> 'nan'  
)  
SELECT Language,  
       CASE  
         WHEN WorkExp::INT <= 2 THEN '0-2 Years'  
         WHEN WorkExp::INT <= 5 THEN '3-5 Years'  
         WHEN WorkExp::INT <= 10 THEN '6-10 Years'  
         WHEN WorkExp::INT <= 15 THEN '11-15 Years'  
         ELSE '15+ Years'  
       END AS Experience_Level,  
       ROUND(PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY Salary)) AS Median_Salary  
FROM Languages  
GROUP BY Language, Experience_Level  
HAVING COUNT(*) > 30  
ORDER BY Median_Salary DESC;
```

5. Key SQL Techniques

Technique	Usage in Project	Purpose
PERCENTILE_CONT(0.5)	All salary queries	Calculate median (robust to outliers)
CASE WHEN	Experience bucketing	Categorize numeric ranges into labels
STRING_TO_ARRAY(col, ';')	Language/DB/Platform queries	Split semicolon-delimited strings into arrays
UNNEST(array)	Language/DB/Platform queries	Expand array into individual rows
CTE (WITH clause)	Technology analysis queries	Modularize multi-step transformations
HAVING COUNT(*) > N	All grouped queries	Filter out statistically unreliable groups
::NUMERIC, ::INT	Throughout	Type-cast TEXT columns for calculations
GROUP BY ... ORDER BY	Throughout	Aggregate and rank results

6. Setup & Execution

6.1 Prerequisites

- PostgreSQL 12 or higher
- psql command-line client
- Developer_survey.csv file in the working directory

6.2 Steps

Step 1: Create the database and table

```
psql -d your_database -f project_stackoverflow.sql
```

Step 2: Import the CSV data

```
psql -d your_database -c "\COPY developer_survey FROM 'Developer_survey.csv'
CSV HEADER"
```

Step 3: Execute individual queries or run the full script

```
psql -d your_database -f project_stackoverflow.sql
```

7. Key Insights

Insight	Finding
Experience & Pay	Median salary scales consistently with years of experience; 15+ year developers earn significantly more than early-career developers.
Remote Work Premium	Remote developers report higher median salaries than hybrid or in-person workers globally.
Top Languages	Certain languages (e.g., Go, Scala, Kotlin) consistently appear at the top of the median salary ranking.
Database Choices	Cloud-native and enterprise databases tend to correlate with higher compensation.
Geographic Variation	U.S.-based developers report some of the highest median salaries; significant variation exists across regions.
Technology Stack	Platforms associated with DevOps, cloud infrastructure, and containerization appear among the highest paying.

8. File Structure

File	Description
Developer_survey.csv	Raw dataset with ~49,200 developer survey responses
project_stackoverflow.sql	Complete SQL script: table creation, data loading, and all analysis queries
README.md	GitHub project documentation and quick-start guide

9. Limitations & Considerations

- Salary data is self-reported and may not reflect actual compensation accurately.
- ConvertedCompYearly represents an approximation based on exchange rates at the time of the survey.
- Multi-value fields (languages, databases) do not indicate primary or preferred tool - only that the developer has worked with them.
- Respondents are self-selected Stack Overflow users, which may skew toward more experienced or tech-forward developers.
- Countries with fewer than 50 respondents are excluded from the country-level analysis to reduce noise.

10. Key Takeaways

- Experience was the strongest compensation driver, with median salaries increasing from \$22,667 (0–2 years) to \$104,413 (15+ years).
- Specialized programming languages such as Ruby, Erlang, Perl, Elixir, and Scala were associated with the highest median salaries.
- Modern cloud and analytics databases, including Snowflake, Amazon Redshift, and Databricks SQL, commanded premium compensation.
- Observability and infrastructure tools such as Splunk, Datadog, Terraform, and New Relic were linked to higher-paying roles.
- Advanced PostgreSQL techniques enabled efficient analysis of large-scale survey data and identification of key compensation trends.

11. Conclusion

This project analysed 49,200 Stack Overflow Developer Survey 2025 responses using PostgreSQL to identify key compensation drivers across experience levels, programming languages, databases, and platforms. The findings revealed that experience, technology specialization, and expertise in modern cloud and data technologies are strongly associated with higher developer salaries. Advanced PostgreSQL techniques enabled efficient transformation and analysis of large-scale survey data, demonstrating the value of SQL for data-driven decision-making.